An Unsupervised Approach for Knowledge Construction Applied to Personal Robots

Cristiano Russo, Kurosh Madani, and Antonio Maria Rinaldi

***Abstract*—The employment of personal robots or service robots has aroused much interest in recent years with an amazing growth of robotics in different domains. Although sophisticated humanoid robots have been developed, much more effort is needed for improving their cognitive capabilities. Interactions with humans and/or with other agents are still limited and not considered satisfactory. So, the way we store and represent knowledge in a cognitive architecture is fundamental in order to overcome these limitations and improve the human–machine and machine–machine interactions. In this article, we propose an unsupervised approach for knowledge construction based on the robot's perception. Our approach makes use of Kohonen maps as an unsupervised machine learning technique and allows the definition of semantic clusters from visual features perceived by the robot. Besides, a multimedia graph knowledge base using a pure formalism is presented, which can be actively used by personal robots in their classic activities, such as environment exploration or information gathering, to represent and share the acquired knowledge, linking it to abstract concepts gifted with semantic relations.**

***Index Terms*—Companion robots, human–robot interaction (HRI), knowledge construction, ontologies.**

I. Introduction

In RECENT years, numerous results have been achieved in the human–computer interaction (HCI) research field. It involves many scientific areas, including artificial intelligence, computer vision, face recognition, motion tracking, etc. However, to truly achieve effective human–computer intelligent interaction (HCII), it is argued that there is a need for the computers to be able to naturally interact with the user, similar to the way human–human interaction takes place [1]. The rapid evolution we have witnessed in recent years in the field of artificial intelligence and robotics has given a great impetus to human–robot interaction (HRI). We are quickly passing from conventional robots to service robots and companion robots with a significant change in the relationship between humans and robots. While the firsts are, from a general point of view, noninteractive and working in safe and isolated environments, the second ones are designed to assist humans in their daily activities and to interact with them. For the next generation of robots, there is the need to enhance the cognitive functionalities and intelligence of human–robot systems for better interaction performance, involving an interdisciplinary research effort coming from robotics, computer science, neuroscience, and psychology. A number of challenges should be addressed urgently. The new generation of robots is expected to be highly intelligent, to learn new skills, and to perform novel tasks with their partners through interaction. Meeting these requirements is not an ordinary task. Humans are able to communicate through multiple modalities. They can interact through speech, writings, display of emotions but also through body gestures defined by established conventions within the same cultural line. In the face-to-face exchange, humans employ these communication paths simultaneously and in combination, using one to complement and enhance another. The exchanged information is largely encapsulated in this natural, multimodal format. But the roles of multiple modalities and their crossing remain to be quantified and scientifically understood [1]. As a consequence, the research community aims to find new interface technologies to make HRI in a highly intelligent and friendly manner, accommodating information exchanges via the natural sensory modes of sight, sound, and touch through multimodal learning techniques for robot cognition development. Although sophisticated humanoid robots have been developed, much more effort is needed for improving their cognitive capabilities and achieve an effective human–robot friendly interaction. In our vision, cognition plays a fundamental role in the development of human-friendly robots and intelligent artificial systems able to interpret the data at their disposal, transforming them in information and lastly knowledge. The combination of semantic-based techniques and computer vision is a necessary step to accomplish this task [2], [3]. In this article, we present a framework for knowledge construction that looks straightforward in this direction. Moreover, one of the strength points of our framework is to be implementable for different domains of interest due to its generality, which does not tie it to a particular system. Besides, the proposed approach is not limited to the task of object classification, but it is based on general knowledge and aims to extend the current knowledge of an artificial intelligent system with new knowledge. It could be the recognition of a new object, the fusion of existing ones, actions to do referred to specific concepts, etc. This article,

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Cristiano Russo and Kurosh Madani are with the LISSI Laboratory (EA 3956), Université Paris-Est, 94000 Créteil, France (e-mail: cristiano.russo@univ-paris-est.fr; madani@u-pec.fr).

Antonio Maria Rinaldi is with the Department of Electrical Engineering and Information Technologies, University of Naples Federico II, 80125 Napoli, Italy, and also with IKNOS-LAB Intelligent and Knowledge Systems (LUPT), University of Naples Federico II, 80125 Napoli, Italy (e-mail: antoniomaria.rinaldi@unina.it).

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in fact, is based on what a system (it may be a robot or any machine as well) perceives from the surrounding environment using an unsupervised approach. The result is a cognitive map containing the semantic concepts which are stored in a general multimedia knowledge base. The proposed approach aims to discover new patterns of knowledge based on the available information retrieved from the surrounding environment. The novelty of the presented method resides in the way it operates to reach its target, that is, the merging of two different well-known worlds, which are ontologies and machine learning techniques. The first one is a valuable tool for knowledge representation and design, but it suffers from rigid rules to follow, which is in contrast with the flexibility requirements of a complex research area such as knowledge construction. The second one can be exploited to fill the aforementioned gaps and to reach better performances in the knowledge construction task, as in the scenario of a friendly HRI. To the best of our knowledge, existing approaches to knowledge construction rarely combines semantics-based techniques with machine learning techniques and, when this happens, they do not use an unsupervised approach, or they pursue many different objectives from the ones discussed above. The remainder of this article is organized as follows. In Section II, we introduce some works related to the knowledge construction field and HRIs. We review works sharing concepts with the purpose of this article, that is, the use of a bio-inspired approach for improving the cognitive skills of machines. In Section III, we present our framework, describing the multimedia semantic knowledge base and the proposed unsupervised approach. Section IV is devoted to the assessment of our methodology through experiments. Conclusions and future research are discussed in Section V.

II. Related Works

HRI is a research field in constant growth, with multiple issues which need to be addressed, from multimodal sensing and perception to design and human factors, and from developmental and epigenetic robotics to social, service and assistive robotics, and robotics for education. An overview of the major research challenges within HRI is presented in [4]. In this section, we mainly focus on research works related to knowledge construction, cognitive capabilities development applied to service robots, which are intended to have friendly interactions with humans. In [5] and [6], a cognitive map is created in order to build friendly interactions between the user and the robot. The authors propose a method to identify the spatial relation between the objects in a given environment and describe such environments using uncertain terms related to spatial information based on a cognitive map created by the proposed system while having interactive conversations with the user. Other works address specific aspects and features that an HRI should have. Recently, much research efforts have been made with particular interest in the field of human–robot collaboration (HRC). It is an interdisciplinary research area comprising classical robotics, cognitive sciences, and psychology. A survey of the state of the art of HRC is given in [7]. Established methods for intention estimation, action planning, joint action, and machine learning are presented together with existing guidelines for hardware design. A more recent survey on HRC can be found in [8]. Zhao and Pan [9] presented a method to accomplish smooth HRC in close proximity by taking into account the humanŠs behavior while planning the robotŠs trajectory. They use an occupancy map to summarize humanŠs movement preference over time and such prior information is then considered in an optimization-based motion planner via two cost items: 1) the avoidance of the workspace previously occupied by a human to eliminate the interruption and 2) to increase the task success rate; the tendency to keep a safe distance between the human and the robot to improve the safety. Podpora and Rozanska [10] described a new approach to the analysis of user experience in interacting with the humanoid robot *Pepper*. In our vision, the human–robot collaboration task is not our primary goal but it is a required activity to support the construction of a robot knowledge conceptualization. Chivarov *et al.* [11] developed human–robot interfaces, designed to provide user-friendly interaction between the elderly or disabled and the robot ROBCO 17. Menne and Schwab [12] investigated the multilevel phenomenon of emotions by using a multimethod approach. Shidujaman *et al.* [13] presented the design, control, and expressive capabilities of *RoboQuin*, a novel humanoid social robot. With respect to our approach, the aforementioned works deal with the theme of interaction at different levels and dimensions, i.e., they focus on robot facial expression or robot body design and dynamic robot postures on human perception. Coronado *et al.* [14] proposed a software framework to develop intelligent behaviors in social robots. The use of ontologies for handling knowledge in robot applications has been very successful. Robot task planning is investigated in [15]. The authors make use of semantic maps, which integrate hierarchical spatial information and semantic knowledge. The OpenRobots Ontology (ORO) [16] is a knowledge management module for cognitive architectures in robotics. It allows to turn previously acquired symbols into concepts linked to each other. KnowRob [17] is a knowledge processing system designed for autonomous personal robots providing tools for action-centered representation and automated acquisition of grounded concepts. Our proposed approach shares some points with the works cited above; in particular, our ontological model has been implemented in a graph knowledge base. However, in our work, we have not a particular focus on action-oriented operations but it is possible to include them in our knowledge base. Instead, our goal is the acquisition of grounded signs through the use of ontologies with the purpose of knowledge conceptualization. Most of the works mentioned, share similar peculiarities, i.e., the study of human behavior or improving HRC, etc., but they are focused on the interaction itself and often the consciousness ability follows a human-driven interaction. Other works [18], [19] combine ontologies and machine learning techniques similar to our approach. However, they have different aims or they are loosely related to the robotic field. The first paper combines SVM and ontologies to extract information for a recommendation system in social robots. The second one is a system for leaf classification in plant identification. In this article, we propose novel methods and technologies to provide a robot or, more in general, an artificial system, with the ability of actively and autonomously developing its cognitive skills prior to interaction, through an unsupervised approach, which is helpful for the definition of a more friendly and intelligent interaction with humans. Moreover, our framework allows a dimensional features reduction which speeds up the computation of the implemented algorithms and it is also based on semantics, which is the keystone in our cognitive development approach. In particular, the main features of this article compared with the presented literature are: 1) realization of a multimedia knowledge base containing linguistic, semantic, and visual representation of general concepts; 2) implementation of an unsupervised approach for knowledge construction by means of self-organizing maps which successfully provided semantically defined conceptual cluster from visual features; and 3) application of the proposed approach to a humanoid robotic platform in a home and office environment.

III. Proposed Framework

The development of artificial intelligent systems, such as service or domestic robots, is one of the current trends in AI. We have seen in Section II how many research studies focused on various application fields, e.g., healthcare, industries, and domestic use to assist people have been carried out. Such a research topic is not trivial due to its intrinsic interdisciplinarity. In order to have satisfactory interactions and knowledge acquisition skills, either with humans or artificial systems, we argue that efficient techniques to learn cognitive process are the ones based on bio-inspired approaches [20]–[23]. Moreover, a solid representation of knowledge and an efficient system for storing the acquired one is required. In the following, we present our method for knowledge conceptualization, then we introduce our approach for its construction.

A. Knowledge Base

A top-level multimedia ontological model is used as a general knowledge base in this article. It is an abstract model based on the work proposed in [24]. This formal representation uses *signs* which represent concepts, defined in [25] as "something that stands for something, to someone in some capacity." Generally speaking, a sign can be any multimedia form, such as text, images, gestures, sounds, or any other form through which knowledge can be conceptualized. Each type of representation has some properties that distinguish them. From a set theory point of view, the model can be seen as a triple $\langle S, P, C\rangle$ where: S: the set of signs; P: the set of properties used to relate signs with concepts; and C: the set of constraints on the P collection.

The basic idea behind this model is to add multimedia capabilities to WordNet [26], a lexico-semantic dictionary for the English language. The properties between concepts are linguistic and semantic relations and the constraints contain validity rules applied to properties with respect to the considered multimedia representation. Knowledge is represented by an ontology implemented using a semantic network (SN). It can be seen as a graph where the nodes are concepts and the arcs are relationships between them. A concept is a set of multimedia data representing an abstract idea. In this article, we make use of two types of representations, one textual, following the WordNet structure, and one *visual* (i.e., images referring to specific concepts). Fig. 1 shows the class hierarchies related to concepts, multimedia, lexical, and semantic properties. The MM class with the respective subclasses represents all the possible signs of our ontology. The classes have no elements in common and therefore are defined as disjointed. Table I shows some of the considered properties. Classes are also provided with attributes. The *Concept* classes have the attributes: *Name* which represents the name of the concept and the field *Glossary* which contains a short description of it. The attributes of the MM class are: *Name* and *ID*. Each subclass has its own set of features depending on the nature of the media. In this article we use textual and visual representation and, in the *Visual* case, we consider various visual descriptors, as attributes of multimedia (MM) nodes. In particular, for each image retrieved from ImageNet, we extract six global features and two "deep" features, taken from activation layers of deep learning models. In this article, we considered the feature vectors taken from the last activation layers of two prominent deep convolutional neural networks (CNNs), the *VGG-16* [27] and *googleNet* [28]. The use of activation features for generic tasks in visual recognition is justified in [29]. The relations in the SN are represented as *ObjectProperties*. They depend on the syntactic category of the considered concept. Some of these properties (semantic or lexical) are shown in Table I.

For example, the hypernymy property can only be used between nouns and nouns or between verbs and verbs. Each multimedia is linked to the represented concept by the ObjectPropety *hasConcept*, *vice versa* with *hasMM*. They are the only properties that can be used to link concepts with multimedia, the remaining properties are used to link

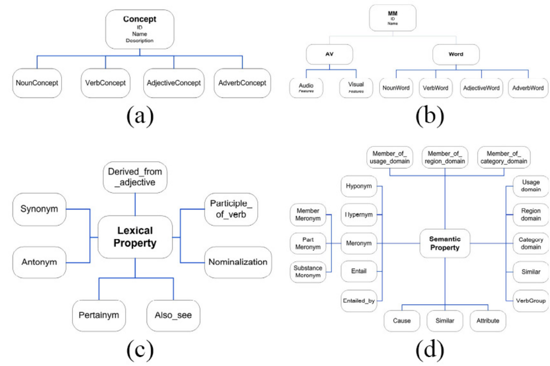


Fig. 1. Concept, multimedia, and linguistic properties defined in the model.

TABLE I
PROPERTIES

Property	Domain	Range
hasMM	Concept	MM
hasConcept	MM	Concept
hypernym	Nouns and Verbs	Nouns and Verbs
holonym	Noun	Noun
Entailment	Verb	Verb
Similar	Adjective	Adjective

TABLE II
PROPERTIES CONSTRAINTS

Constraint	Class	Property	Constraint range
AllValuesFrom	Noun	Hypernym	Noun
AllValuesFrom	Adjective	attribute	Noun
AllValuesFrom	Verb	also see	Verb
AllValuesFrom	Noun	Hyponym	Noun

TABLE III
PROPERTIES FEATURES

Property	Features
hasMM	Inverse of hasConcept
hasConcept	Inverse of hasMM
hyponym	Inverse of hypernym: transitivity
hypernym	Inverse of hyponym: transitivity
verbGroup	Symmetry and transitivity

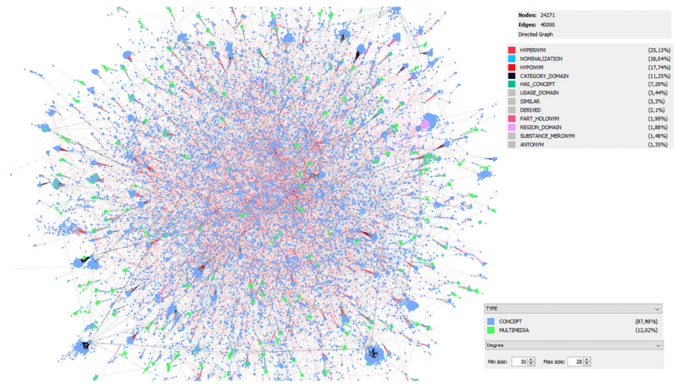


Fig. 2. Excerpt of a macro visualization with Gephi of the knowledge base implemented as a Neo4j instance.

multimedia to multimedia or concept to concepts. The use of union classes gives a precise meaning of properties range of application by means of domain and codomain but at the same time does not exhibit perfect behavior in some cases. For example, the model does not know that a hyponymy property defined on sets of nouns and verbs would have a range of nouns when applied to a set of nouns and a range of verbs when applied to a set of verbs. To solve this problem, we added some constraints and some of them are shown in Table II.

In some cases, the existence of a property between two or more individuals entails the existence of other properties. For example, if a concept A is a hyponym of a concept B, concept B is a hypernym of A. These characteristics are represented in OWL by means of property features. Some examples are given in Table III.

Also, from a robotic side, having a formalized general knowledge base is a valuable and helping tool that may be exploited for many activities carried out by the robot, such as for example, exploration of the environment, information gathering and retrieval, etc. The proposed model has been implemented in a Neo4J GraphDB instance to handle the complexity of semantic multimedia big data.

B. Knowledge Base Implementation

The knowledge base used in our work has been implemented through the ontology model explained in Section III-A. The graph has been designed using a property-based graph model and it is populated with data from WordNet and ImageNet. It defines and describes the concepts expressed by words and makes use of groupings of terms with a similar meaning, called synonym set (synset). Data information in WordNet is highly connected through various types of semantic and lexical relationships. In addition, we use *ImageNet* [30] for the visual representations. Therefore, the proposed ontology can be considered as a graph in which entities (concepts and multimedia) correspond to nodes and ObjectProperties to relations between them. In our system, we use Neo4j as the database system to implement our graph, since it respects the data model of a graph db, it supports *ACID* rules, and it is possible to make queries through a very powerful declarative language, called *Cypher* similar to SQL. Given the huge size of ImageNet and storage limitations, for the purposes of this article, we retrieved up to 15 images for each concept. While such a configuration could slightly affect performances, it is still a valid choice for many reasons. First, it allows us to have better control during the task of the experiment. Another advantage is the balance achieved for the huge number of classes. In fact, one of the drawbacks of ImageNet is its highly unbalance (the number of images per concept may vary significantly). In addition, only about 25% of the WordNet nouns is provided with images. This reduction in terms of computational costs and storage does not prevent us from putting in evidence the complexity of our knowledge base from both dimensional and heterogeneity points of view. In general, we refer to *macro visualization* when we want to graphically represent the whole network or big slices of it, showing all nodes and interconnections between them, highlighting generic aspects regarding topology and complexity. On the other hand, when using the term *micro visualization*, we refer to a small graph with a limited number of nodes for recognizing specific or local aspects of a particular topological area in the network. In Fig. 2, a part of the implemented network is drawn, limited to 24 271 nodes and 40 000 relationships generated by a Cypher query that returns random paths. We have used *Force Atlas 2* as a layout for this figure. In the following figure, the resulting graph is shown together with the statistics offered by the tool. The figure allows us to underline the complexity of the network in terms of the number of nodes and the high number of interconnections. From the statistics offered by the tool and the used colors, it is possible to notice how the hypernymy relationship outclasses the others.

In Fig. 3, the number of nodes has been limited in order to display only the concepts directly connected by a single type of relationship. In this case, we use the standard graphic Neo4j interface. From this graph, we can highlight, for example, how the ontology rules and constraints defined in Section III-A are respected (inverse relation to another or constraints of type *allValuesFrom*). We can observe from the network topology that starting from any concept it is possible to retrieve not only the concepts directly connected to it but also other semantic links such as the hierarchical path composed of hypernymy relationships and hyponymy to the root. We also improved

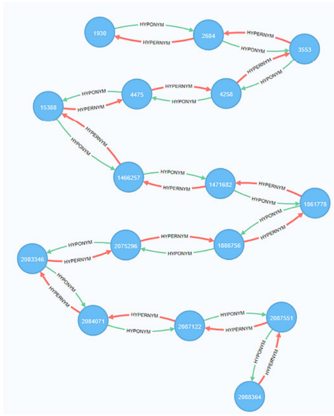
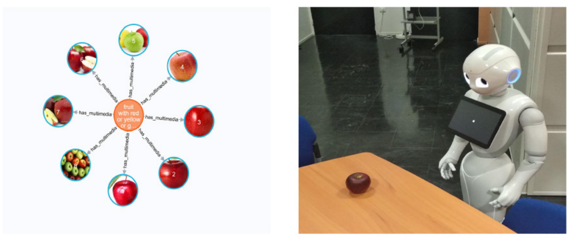


Fig. 3. Micro visualization of an SN highlighting the hyponym-hypernym pairs.



(a) (b)

Fig. 4. On the left, the concept apple is stored in the knowledge base, along with its multimedia items. On the right, the robot Pepper acquiring the same object in a real scenario. (a) Visualization of the concept apple and the related multimedia elements. (b) Acquisition of the concept from the robot view.

the visualization of the knowledge base by adding a third-party javascript extension to the Web application provided by Neo4j. It is used to represent the visual nodes with the respective original images. The major benefit, apart from a better *look and feel* when displaying subgraphs of the network, is more complete control over the knowledge base contents when testing the approach. In order to provide an overview from both the knowledge base side and the robotic one, we provide in Fig. 4 an example of how the knowledge is conceptualized and acquired. Fig. 4(a) shows how the concept "apple" is stored with its associated visual representations.

Fig. 4(b) shows the humanoid robot *Pepper* committed to acquire the same concept during one of our experiments in a real environment. Pepper is designed to be fully interactive, as it can hold full conversations, and it is endowed with a number of sensors, allowing it to speak, detect obstacles, take photos/videos, move, and so on. The robot is equipped with a speech recognition engine which provides the capabilities of translating to text what a human says. In particular, for visual sensors, it is equipped with two 2-D cameras and an ASUS Xtion 3-D sensor.

C. Proposed Knowledge Construction Approach

Once we have set-up our semantic multimedia knowledge base, we can move on describing our approach for constructing the knowledge. More specifically, our focus is on the development of bio-inspired approaches based on artificial cognition for robots' autonomous acquisition of knowledge. It must be said, however, that the core of our approach is still valid with application domains different than robotics. The main aim is to allow the robot to extract, to construct, and to conceptualize the knowledge with a dual process, combining together perceptual knowledge and semantic knowledge. The perceptual knowledge comes from the surrounding environment. It generally embraces all the information retrieved from sensors, e.g., vision for extracting images, microphones for capturing audio signals, such as human conversations, movements, temperatures, and so on. In this article, we will visual sensors because they deal with information extracted from images. The data acquisition process can be extended, in order to acquire knowledge from other sensors, such us microphone for audio, infrared for distances, etc. The semantic knowledge is the one represented and implemented in our knowledge base. The validation of investigated concepts will be carried out on a humanoid robotic platform (see Section IV). The dual process above-cited looks into two different directions. The goal is to grasp the advantages of the two methods. The first is more abstract, it provides semantics and it is closer to human understanding. The semantic Web technologies, ontologies, linked open data, production rules systems, etc., fall into this category. The second one is closer to machine understanding and it is related to perception, raw information extracted from sensors. A first approach has been presented in [31]. It is a *top-down* approach, meaning that the main flow of the process relies on semantic knowledge. In this article, we present a *bottom-up* approach, more related to perceptual knowledge. Starting from a collection of images representing a subset of WordNet synsets, our goal is to use unsupervised learning methods to check how resulting visual clusters are dislocated, if they follow the line of semantic knowledge and if they could be exploited for discovering new concepts and/or establishing new relations between existing concepts. The choice of using these kinds of methods naturally comes from the assumption for the bottom-up approach to not have any a priori knowledge for the visual information. Unsupervised learning helps to find unknown patterns in data sets without preexisting labels. One particularly interesting class of unsupervised system is based on competitive learning, in which the output neurons compete amongst themselves to be activated, with the result that only one is activated at any one time. This activated neuron is called a *winner-takes-all neuron* or the winning neuron. The result is that the neurons are forced to organize themselves. Such a network is called a self-organizing map (SOM) [32]. During the training process, the neurons become selectively tuned to various input patterns (stimuli) or classes of input patterns during the course of the competitive learning. The update formula for a neuron v with weight vector $W_v(i)$ is

$$W_v(i+1) = W_v(i) + \theta(u, v, i) \cdot \alpha(i) \cdot (D(t) - W_v(i)) \quad (1)$$

where i is the iteration index, t is an index into the training sample, u is the index of the best matching unit (BMU) for the input vector $D(t)$, $\alpha(i)$ is a monotonically decreasing learning coefficient; and $\theta(u, v, i)$ is the neighborhood function which gives the distance between the neuron u and the neuron v in step i. The neighborhood function depends on the grid distance

TABLE IV
LISTS OF CONCEPTS FOR EACH SEMANTIC DOMAIN

Semantic Domain	Synsets
Animals	cat, bird, dog, elephant, giraffe, zebra, sheep, cow, horse, bear
Food	banana, apple, sandwich, orange, broccoli, hotdog, pizza, watermelon, pear, donut, cake
Vehicles	car, truck, boat, airplane, bicycle, motorcycle, bus, train
Dishes	spoon, cup, knife
Other	person, potted plant

between the BMU (neuron u) and neuron v. Regardless of the functional form, the neighborhood function shrinks with time. From a robotic point of view, the used approach is applicable to a humanoid robot to evolve and improve its knowledge about the environment in order to assist humans through more friendly interactions (see Section IV-B).

IV. EXPERIMENTS

In this section, we show a set of experiments we carried out in an *ad hoc* test environment to answer the previous questions and give an evaluation of our approach. Given the nature of our context, we will use in our analysis both qualitative and quantitative measures. In fact, our purpose is to focus on some specific skills that the system should have for a knowledge construction task. In particular, the autonomy level for the acquisition of knowledge and the evolving ability.

In the first step of our experiments, we used a data set with 34 synsets, manually selected from our knowledge base to have wide coverage of general knowledge. More specifically, the selected concepts may be grouped into four semantic domains. Table IV lists all the synsets used in our experiments grouped by the semantic domain.

We defined five semantic groups of classes. The first four are well-defined from a conceptual point of view, while the last one has been intentionally filled with two different classes, including plants and persons which do not belong to the same semantic domain. As we pointed out in Section III-B, despite the huge size of our knowledge base, we retrieved a limited number of images for each synset. For this reason, we tried to overcome this limitation for our experiment extending the number of images retrieved for the chosen synsets. To accomplish this task, we used an automatic script, written in Python language, for image retrieval from the service *google images*. Given the confined number of classes, we also performed a manual check on the retrieved images, discarding any possible inconsistency. The total number of samples is 1477. Detailed statistics about each concept for the data set used in our experiments are provided in Table V.

Samples taken from the data set for some classes of different semantic domains are shown in Fig. 5.

For each added image, all the features described in Section III were extracted and used as input in our experiments.

A. Evaluation

For the set of experiments, we have a number of parameters for the input and for the SOM which need to be tuned.

TABLE V
EXAMPLE OF DATA SET STATISTICS

Class	Num. of images
banana	43
horse	39
bear	38
boat	39
truck	38
spoon	44
train	50
apple	19
person	37
car	40
cup	47
zebra	47
orange	40
pizza	48
elephant	50
bus	49
cat	38
knife	45
airplane	37
dog	37
watermelon	45

Fig. 5. Sample images for classes airplane, bear, pizza, and spoon.

In particular, we have ten different choices for the input features (six global features + two activation features). The other parameters refer to the SOM configuration. The tuning can be performed by changing the number of iteration for training, the grid size, and the neighborhood function which affects the number of neurons that entail a weights update at each iteration. Vesanto [33] suggested that the maximal grid size for the SOM should be

$$5 * \sqrt{N} \tag{2}$$

where N is the size of the data set. In our first experiments, we tested the global features for different parameter configurations of the SOM. In all figures related to SOM results, each color is mapped with a semantic domain. The background color depicts the distance of a neuron from its neighborhood.

Global descriptors are in general compact vectors, which take into account colors, textures, and/or shapes of the image. As a consequence, the advantage is that they require much less storage space with respect to features extracted from deep models. On the other hand, for their "global" nature, these vectors take into account all the image content. The peddling information from background or secondary elements in the image is equally processed and put into the final descriptor. This behavior is not desirable for tasks like ours, where the vectors are associated with semantic labels, i.e., the concepts represented in the image. In fact, global descriptors lack the ability of being expressively significant to show a semantic

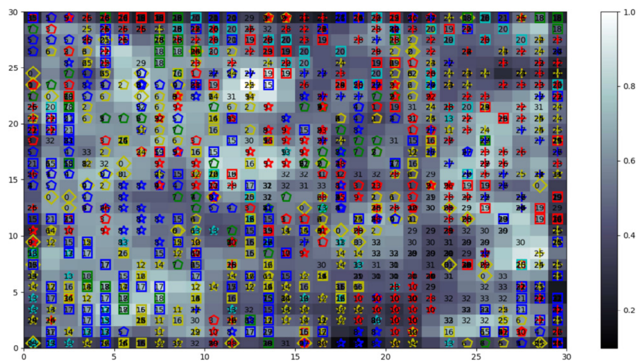


Fig. 6. Results for an SOM with grid size 30 × 30 using JCD features.

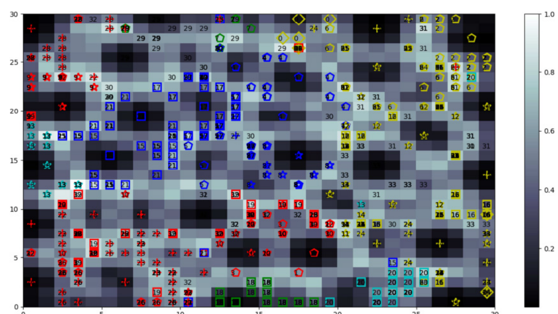


Fig. 7. Results for an SOM with grid size 30 × 30 using VGG-16 features.

relatedness between the concepts. Our expectations have been confirmed by an experiment shown in Fig. 6 for an SOM with grid size 30 × 30 using *JCD* features as input. In fact, global features do not show significant results. Clusters are not very clear, however, we can see that different samples of the same concept are closer with respect to other concepts in some cases. The obtained performances are not to be considered competitive in comparison with more complex image descriptors, like for example the ones extracted from inner activation layers of famous models for CNNs as discussed below. Fig. 7 shows the results obtained for an experiment using the feature vector from the last activation layer (pool-5) of the *VGG-16* CNN. The value of parameters chosen for the SOM are: 30 × 30 grid size; 100.000 iterations; *gaussian neighborhood function* of type

$$e^{-\frac{\|(r_c - r_i)^2\|}{2\sigma^2(t)}}. \tag{3}$$

We used the same color for the synsets belonging to the same semantic domain and different markers symbols for classes. Red is for animals, blue for vehicles, yellow for food, green for dishes, and cyan for others. The background color shows the average distance of the neuron from its neighborhood. The use of colors helps us in the visual inspection of the map. The results obtained for deep feature descriptors as input to our SOM show a significant improvement. In fact, we can see the results show the formation of meaningful semantic clusters. Also for the cyan-colored cluster in which we have grouped people and plants, we can notice that the classes are correctly separated and distant as expected. The semantic proximity of concepts holds in the two-dimensional representation space of the map. The original features vector has 25 088 dimensions.

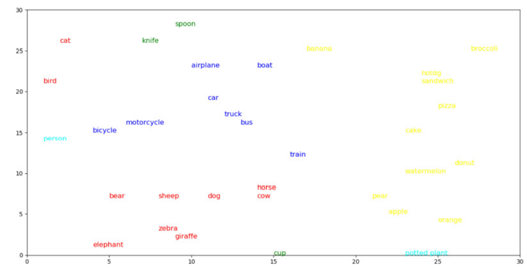


Fig. 8. Positions of centers of gravity in the grid with labels.

TABLE VI

CENTER OF GRAVITY POSITIONS AND WINNING NEURON POSITION FOR CENTER OF GRAVITY VECTOR

Class	Center of gravity position	winning neuron position
banana	(17,25)	(17,26)
horse	(14,8)	(18,8)
bear	(5,7)	(3,4)
boat	(15,23)	(15,25)
truck	(13,17)	(14,16)
spoon	(10,28)	(13,29)
train	(17,12)	(15,13)
apple	(22,6)	(21,8)
person	(1,15)	(0,16)
car	(12,19)	(12,19)
cup	(15,1)	(15,2)
zebra	(8,3)	(8,3)
orange	(25,4)	(26,3)
pizza	(25,18)	(26,20)
elephant	(6,2)	(2,3)
bus	(13,16)	(14,16)
cat	(2,26)	(3,25)
knife	(7,27)	(8,27)
airplane	(11,24)	(11,23)
dog	(11,8)	(13,8)
watermelon	(24,10)	(20,13)

While a qualitative measure like visual inspection is a valuable technique giving fast feedback about the spatial distributions in the SOM, it must be noted that the number of occurrences for winning neurons is not displayed, i.e., markers overlap on the same position.

Based on these results, it is possible to provide quantitative measures for clusters and to compute centers of gravity for each concept. The center of gravity is the average position of all points; furthermore it may not correspond to any of those used for its calculation. In our case, it can be seen as a representative vector of a specific class. It has been computed both in the original 25088-D feature space and in the new 2-D space, see

$$\text{Bar}_c = \frac{\sum_{i=1}^{N} W_i}{N}. \tag{4}$$

Given N feature vectors referring to the concept c, the center of gravity of c is the arithmetic average of the winning neurons. Weight vectors of winning neurons are used to compute the center of gravity in the original feature space, cartesian coordinates of winning neurons (position of the neuron in the grid) for the output 2-D space. The positions of centers of gravity are displayed on the grid in Fig. 8. The numerical results are also provided in Table VI, where we report the centers of gravity coordinates for each class and the winning neuron coordinates for the computed centers of gravity.

The precision of the used features is also evidenced by the fact that similar elements in their appearance, such as for example, motorcycles and bicycles, cows and horses, and zebras and giraffes are very close in the map. Having the

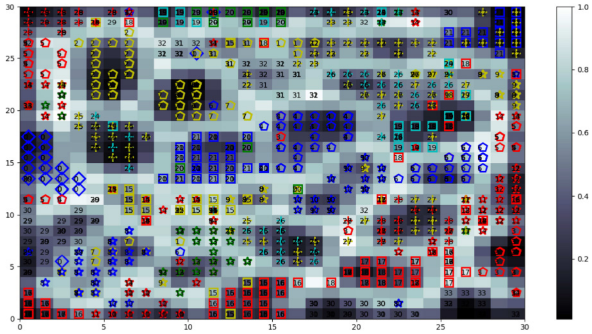


Fig. 9. Results for an SOM with grid size 30×30 using googleNet features.

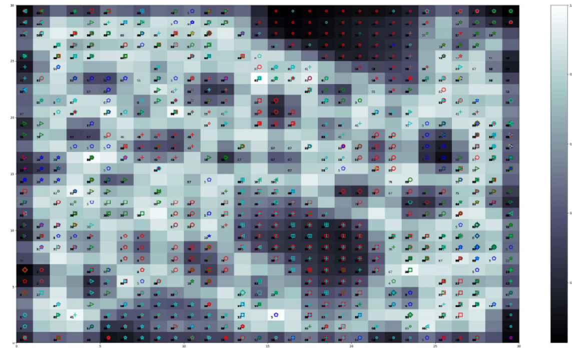


Fig. 10. Results for an SOM with grid size 30×30 using googleNet features (caltech-101).

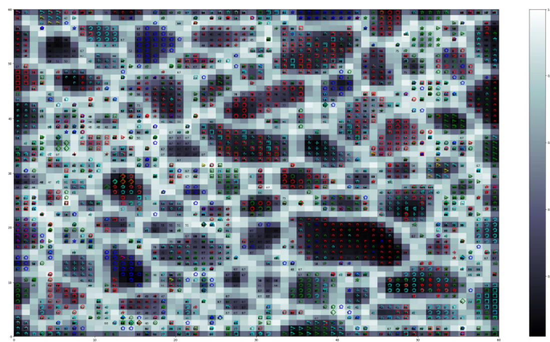


Fig. 11. Results for an SOM with grid size 60×60 using googleNet features (caltech-101).

centers of gravity positions enable us to use a common distance, such as the Euclidean distance, for computing a new semantic distance between the concepts represented in a 2-D space, which is much faster than computing it in a very complex feature space. From the point of view of constructing the knowledge, it is possible for a machine, such as a robot, having at its disposal only the information extracted from its visual field, trying to understand itself (autonomously, without external help), or at least give insights about the latter's belonging to a specific semantic domain, even if the machine has never been trained on that concept before. Moreover, the semantic distance resulting from the unsupervised approach could be actively used in combination with a top-down semantic-guided approach to improve the overall precision of the system. To increase the reliability and significance of our experiments, we have tried different parameter configurations for the features and the SOM. In the experiment shown in Fig. 9, we tested the second considered "deep" feature. It is extracted from the DNN *googleNet*. We used the same configuration for the SOM parameters in the previous "deep" feature to have a comparative measure between the two descriptors. We can notice that the clusters are well-defined at a class-level, i.e., samples of the same class are grouped together in the map. However, there is more sparsity considering semantically related classes. For example, there is a slight overlap for classes of semantic clusters representing the *food* domain (yellow color) and the *vehicles* domain (blue color). To have a more reliable and general testing of our approach, avoiding a dependence of the results from our data set, we tested it on the well-known standard data set *caltech-101*. In order to have a line of comparison with the previous experiments, we inspected the classes of the caltech-101 data set and we manually mapped them to corresponding semantic domain following the structure defined in Table IV (for example, we added the *ant* class to animals, we enlarged the *Dishes* cluster to contain common objects, and so on). Fig. 10 shows the result of an experiment with the same configuration that was used for the previous experiments, both for features (from googleNet) and for parameters of the SOM. The visual inspection suggests that the semantic spatial relationships are somehow still present, despite less clear. Thereby, in the subsequent experiment, we used the same setting for the type of features and the data set, but this time using a different parameter configuration. According to (2) (N for caltech is 9145), we have that the recommended size for the grid should be around 478 neurons, which means around 22×22 neurons. However, the number of overlaps suggests to change the size of the grid (due to the greater size of the caltech-101 data set). For this experiment, the grid size is set to 60×60. As we can see from Fig. 11, the clusters are more sharp, but some semantic relation at domain-level seems to be lost. From this experiment, we can argue that the size of the grid is a point confirming our expectations. All these results are promising and show that it is essential to investigate deeper the approach, with fine tuning of the parameters of the neural network, such as the neighboring function, the number of iteration, the data set used, and hence the quality of images, etc.

B. Implementation on Real Robotic Platform

Based on the results obtained from previous experiments, we used for the experiments with the robot the same setting as shown in Fig. 7, i.e., features extracted from the VGG-16 deep CNN, an SOM trained for 100.000 epochs on the *34-synsets* data set, with a grid size 30×30. The robot is asked to recognize what is in front of him, e.g., an apple [see Fig. 4(b)]. The result of the task execution is shown in Fig. 12. As we can see, the yellow marker shows that the winning neuron for the robot observation is in position $(21, 8)$, which is exactly the same as the one of the computed vector for the center of gravity of the concept apple that is in position $(22, 6)$. In a second experiment with the robot, we went to test his ability to perceive the semantic belonging of a concept represented by an object never seen before, but similar to another

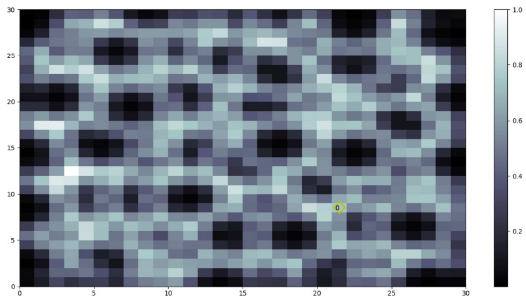


Fig. 12. Results for the apple with the robot Pepper.

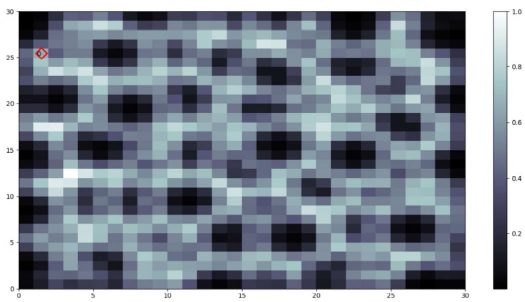


Fig. 13. Results for the Egyptian cat with the robot Pepper.

of which he has already acquired knowledge. To accomplish this task, we want the robot to recognize an *egyptian cat*. The robot, of course, is not able to distinguish the breed of the cat, since it has no information about it. However, the robot *knows* the concept *cat*. So we expect that Pepper is able to recognize at least that it is in front of an animal, or better, a cat. Since we do not have an Egyptian cat in our laboratory, we deceive the robot by showing it an image that represents the cat on a screen (https://en.wikipedia.org/wiki/Egyptian_Mau), simulating its presence.

The winning neuron for the observation (red marker in Fig. 13) is in position $(1, 25)$. The closest center of gravity is that of concept cat [position $(2, 26)$, Euclidean distance $\sqrt{2}$], while the second closest concept is bird, whose center of gravity is in position $(1, 22)$ with a distance set to 3. The robot is therefore led to conclude that it is in front of a cat or some very similar concept. In the context of our approach, the semantic distance defined can be exploited to enable the robot starting an interaction with human or combining the results with a semantic-guided approach to firmly conceptualize the new observation.

V. Conclusion

Representation, design, and conceptualization of knowledge is a demanding and rising task for the new generation of artificial intelligent systems. This article presented a novel unsupervised approach using well-known techniques in machine learning, such as SOMs and more recent CNNs. We also presented a multimedia semantic knowledge base, following a top-level ontological model, based on WordNet and ImageNet. A data set of selected concepts from our knowledge base has been used in the experiments. Features extracted from last activation layers of common CNN models have shown their top-level performances in recognizing objects and their abilities to be adaptable to different uses as generic features. Experiments have shown that the use of SOM, a classic unsupervised neural network, allows to reduce the representation space from 25 088 dimensions (in case of VGG16 features) to just two dimensions while preserving the semantic relations found in the original space. We also computed centers of gravity for concepts, allowing us to interpret the Euclidean distance as a new semantic distance between the concepts represented in the new 2-D space. The dimensionality reduction presents many benefits, in particular, in terms of speeds in calculating distances. Moreover, it is interpretable by humans. The experiments carried out on a different data set and with different features confirm the potential of the approach and highlight some issues that need to be addressed and further investigated. More experiments, with different configurations of parameters, will be conducted. The results obtained are promising and provide some insights that are worthy of interest for future research. First, the use of different sensors for low-level knowledge, e.g., audio-descriptors, to be associated with the perceived image (visual descriptor) to improve the recognition results of the synsets represented in the scene. Then, we will look for other unsupervised strategies to be compared with the proposed one. Another line of research is to evaluate the use of centers of gravity as a representative vector of specific concepts and eventually build an ontology of features based on it. For the knowledge construction and conceptualization task, the use of other approaches as growing hierarchical maps [34], instead of SOM with a fixed size, could be an interesting research line to follow. The strength of such a technique is that of being a naturally evolving approach that could allow us to combine perceptual knowledge with semantic knowledge in a unified model. We also plan to carry out a comparison with other approaches for the object classification task. Moreover, the integration in our knowledge base of domain ontologies following novel methodologies [35], [36] represents an important improvement of the available knowledge represented by our framework.

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Cristiano Russo received the M.Sc. degree from the Department of Electrical Engineering and Information Technologies, University of Naples Federico II, Napoli, Italy, in 2017. He is currently pursuing the Ph.D. degree in joint supervision between Université Paris-Est, Créteil, France and the University of Naples Federico II.

His current research interests include knowledge management, ontologies, information retrieval, machine learning, big data, and data mining.

Kurosh Madani received the graduation degree in fundamental physics from Paris 7-Jussieu University, Paris, France, in June 1985, the M.Sc. degree in microelectronics and the Ph.D. degree in electrical engineering and computer sciences from the University of Paris-Sud, Orsay, France, in September 1986 and February 1990, respectively, and the DHDR Doctor Hab. degree (senior research doctorate degree) from the University Paris 12 Val de Marne, Créteil, France, in 1995.

He works as the Chair Professor of electrical engineering with the Senart-FB Institute of Technology, University Paris-Est Creteil, Créteil, where he was the Co-Creator with the Images, Signals, and Intelligent Systems Laboratory (LISSI/EA 3956) in 2005. He has authored and coauthored of more than 350 publications (in international journals, books, conferences', and symposiums' proceedings).

Dr. Madani has been awarded by the Academic Palm of French Government (Chevalier de l'Ordre des Palmes Académiques) for his academic and scientific achievements in 2017.

Antonio Maria Rinaldi received the graduation degree in computer engineering and the Ph.D. degree in computer science and systems from the University of Napoli Federico II, Napoli, Italy, in 2005, with a dissertation on the use of ontologies and semantic relatedness for information retrieval on the Web.

In the last years, he has been involved in several research and industrial projects related to information retrieval, multimedia database integration, and geographic information systems. He is currently an Assistant Professor with the Dipartimento di Ingegneria Elettrica e delle Tecnologie dell'Informazione, Napoli. He is the Director of the IKNOS-LAB Intelligent and Knowledge Systems (LUPT), University of Napoli Federico II. His fields of interests are knowledge representation and retrieval, big data, multimedia database, and GIS.